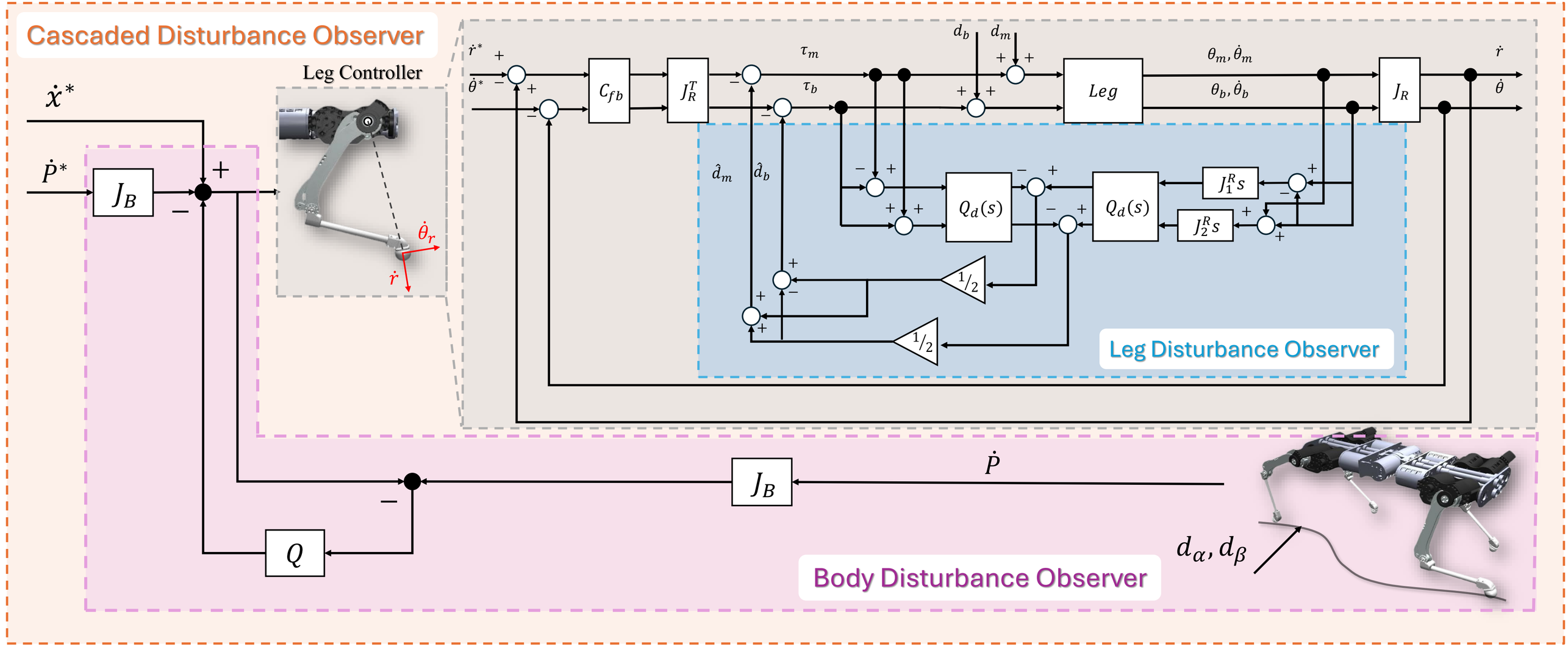

Cascaded Disturbance Observer
Leg Controller
$\dot{x}^*$
$\dot{P}^*$
J_B
$\dot{\theta}_r$
$\dot{r}$
$\dot{r}^*$
$\dot{\theta}^*$
C_{fb}
J_R^T
τ_m
τ_b
d_b
d_m
Leg
$\theta_m, \dot{\theta}_m$
$\theta_b, \dot{\theta}_b$
J_R
$\dot{r}$
$\dot{\theta}$
$\hat{d}_m$
$\hat{d}_b$
$Q_d(s)$
$Q_d(s)$
$J_1^R s$
$J_2^R s$
$1/2$
$1/2$
Leg Disturbance Observer
J_B
$\dot{P}$
Q
d_α, d_β
Body Disturbance Observer